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Revitalizing Borealis Wealth Fund

Aligning Finance with Biodiversity and Net-Zero Goals



Nuuksio National Park, Finland



SQUARE CAPITAL

A strategic proposal for our dual-mandate sovereign wealth fund focused on restoring Finland's forests, protecting biodiversity, and delivering sustainable financial returns.

EK SQUARE CAPITAL

The Imperative to Realign BWF's Strategy

Time is running out. We are only ten years away from Finland's national goal of reaching net-zero emissions, yet the horizon looks bleak. Unsustainable forestry practices and rising temperatures have turned Finnish forests from a carbon sink into a carbon source. Peatland loss and rising carbon emissions now threaten Finland's climate targets, the long-term viability of its forestry sector, and the overall health of its ecosystems.

Our sovereign fund was established to advance ecological goals, but this is no longer sufficient. Finland now faces a far greater challenge, and the existing framework lacks the scope to address it. Immediate action is needed to realign our investments and restore both ecological and economic value.

Mounting pressure from the EU, through its biodiversity strategy and regulations such as LULUCF, adds urgency. This brings higher costs from limits and compliance, but also opportunities through subsidies, green financing, carbon and biodiversity credits, and stronger long-term returns. The trade-off is short-term margin pressure in exchange for lower risk, higher value, and access to ESG-focused markets.

Our New Dual Mandate Policy

The Borealis Wealth Fund is committing to a long-term strategy that balances biodiversity preservation with financial sustainability. Our new motto is Green Growth, Lasting Returns. With ten billion euros in assets, we will actively reshape Finland's forestry sector and restore ecosystems. We will align the objectives of all stakeholders and direct investments to preserve biodiversity in Finland and beyond, ensuring our actions deliver both ecological and financial impact.

BWF

Active Stewardship and Engagement

The work begins today, guided by a clear plan to achieve our objectives. We will not rely solely on screening investments or providing passive financing. To preserve forests, restore peatlands, and reduce carbon emissions without compromising economic value, we will take an active role. We will engage directly with companies to ensure all actions are aligned with our goals.

Aligning Financial and Ecological Goals

We aim to balance financial returns with biodiversity impact. By analyzing both metrics together, we identify investments that deliver strong performance while supporting ecological goals. The goal is a portfolio where one objective does not compromise the other, ensuring the Borealis Wealth Fund advances economic and environmental value simultaneously.

Taking a Stake Where it Matters

We focus investments in Finland to support local forestry and related sectors. This allows us to engage directly as active shareholders, guide sustainable practices, grow returns, and allocate funds to forest conservation while managing risk. We also invest globally to attract capital and gather expertise for Finland, creating compounded ecological benefits for both the economy and the environment.



Decrease
annual
deforestation



Higher long term
returns
expected



Protect 35% of
Finland timber
products



Biodiversity
protection in
forests



Climate
adaptation and
management



Compliance
with Finland's
regulation

About Borealis Wealth Fund

The Borealis Wealth Fund (BWF), founded in 2021, is Finland's long-term investment vehicle, designed to strengthen the ecological resilience of the country's forests while delivering stable financial returns.

While BWF has primarily focused on achieving ecological and financial targets, it is now fully committed to a dual mandate of biodiversity preservation and financial sustainability, aligned with Finland's 2035 net-zero goal and the COP15 30x30 target.

Linked directly to the performance of the forestry sector, BWF ensures economically sustainable investments that generate measurable ecological impact, operating independently of regulatory authority.

Forestry at the Crossroads of Economy and Ecology

Finland's forests cover over three-quarters of the country and are central to its economy, exports, and jobs.¹ The forestry sector alone supports tens of thousands of workers and underpins a wider bioeconomy that includes pulp, paper, bioenergy, and biomaterials. Forests are also a source of innovation, from carbon-storing building materials to bio-based chemicals. With global demand for sustainable products rising, we must find a solution to balance economic growth with protecting biodiversity and maintaining carbon storage.

Threats and Rules Shaping Finland's Forests

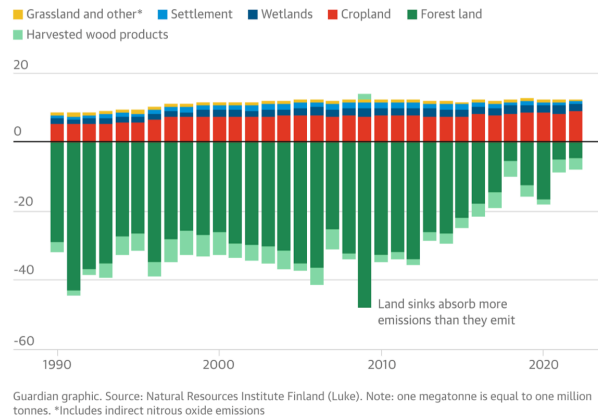
Old-growth forests are shrinking, and fast-growing monocultures reduce ecosystem diversity and resilience. Climate change, soil emissions, and intensive harvesting have even turned forests into a net carbon source, challenging Finland's 2035 net-zero target. Nearly 30% of forest species are now threatened.²

Programs like METSO, Helmi Habitat, and Priodiversity LIFE help protect and restore forests, but funding is

limited. BWF can step in to invest in sustainable forestry and restoration projects, linking financial returns to ecological outcomes and helping safeguard both biodiversity and Finland's long-term economic value.

Finland's land sink has begun to release more emissions than it absorbs

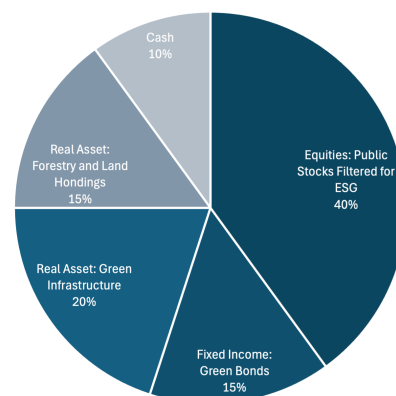
Emissions from land use, land-use change and forestry, megatonnes of CO₂ equivalent



Our Current Holdings

Over the past four years, the Fund has achieved an annualized return of 6.5% with an average volatility of 11.27% and a Sharpe ratio of 0.24.

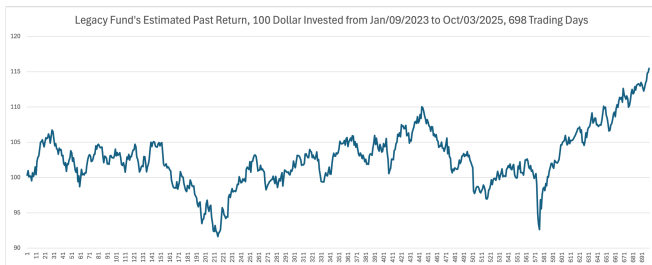
Legacy Portfolio Composition



As seen in the above diagram, 40% of the portfolio was invested in equities, primarily in Finland, with selective global exposure for diversification. The

remainder was allocated to green financing, real assets, and a cash buffer.

ESG equities led performance with a 14.3% annualized return, while fixed income contributed stability at 1.7%. Forestry returned 0.5% and green infrastructure fell 6.7%, both maintaining strong long-term potential. A 10% cash buffer helped manage volatility without reducing overall returns.



The Fund showed resilience through mid-2023 drawdowns and rebounded strongly into 2025. Gains reflect improving sentiment in sustainable assets, EU policy support, and disciplined portfolio construction balancing growth and defense.

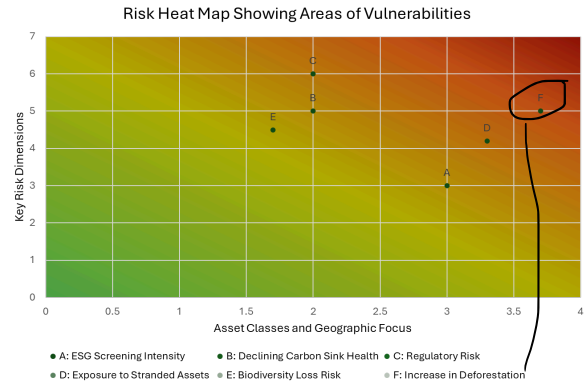
Our Current Strategy and Its Limitations

Our fund has until now relied on stringent ESG screening. Our process involved applying baseline exclusions to industries that violate ESG norms and focused on firm-level metrics such as GHG emissions, water use, waste management, and social and governance standards. Most green bond and forestry investments were screened similarly, meeting broad sustainability criteria but lacking binding biodiversity commitments. This made the portfolio “green” in appearance, yet it delivered limited measurable benefits for species, habitats, or ecosystems.

Screening alone no longer meets the Fund’s dual mandate. Firm-level ESG metrics do not capture landscape-level impacts, such as habitat fragmentation, soil degradation, or ecosystem health. Simply avoiding certain companies may make the portfolio look responsible on paper but does little to improve ecological outcomes. Biodiversity recovery

requires capital and active stewardship in sectors where change is most needed.

Rapid regulatory and ecological shifts in forestry and related industries also pose tangible risks to asset value. Relying on static, backward-looking ESG scores leaves the Fund slow to respond to emerging risks and ecological tipping points.



Considerations Moving Forward

To deliver real impact, BWF must move beyond exclusion and engage directly with companies, projects, and sectors. The updated strategy will prioritize investments that actively regenerate ecosystems, create measurable biodiversity outcomes, and strengthen long-term financial resilience.

Aligning Investments with Net-Zero and Biodiversity Goals

Our first priority is to use our investments to help Finland reach its 2035 net-zero target while preserving biodiversity. Our estimation shows that that would require on average a reduction in GHG emissions by roughly 30 percent annually (see Appendix), which is no small task. Policies must integrate forestry and biodiversity strategies focusing on resilience, adaptation, and ecosystem-based approaches rather than single-goal optimization.

Key actions include restoring degraded ecosystems, enhancing old-growth forests, carefully introducing exotic species to maximize carbon capture without harming biodiversity, and reducing harvesting intensity

to protect carbon sinks. These measures must balance ecological impact with economic considerations.

Adaptive Forest Management for Long-Term Value

Current forestry practices, including clear-cutting and monocultures, maximize output and support most of Finland's exports. However, these methods degrade biodiversity, reduce old-growth forest area, and weaken ecosystem resilience. Shifting toward selective logging and mixed-age native forests may reduce short-term output by 15 to 30 percent, but secures long-term ecological and economic value.

A mosaic approach will combine high-yield plantations, where impacts can be managed, with zones of old-growth and mixed forests. Emphasizing value-added and premium products preserves profitability while keeping carbon stored in wood. Temporary reductions in output are necessary but can be managed to maintain trade relationships and long-term export potential.

Strategic Diversification Beyond Finland

Supporting local biodiversity is essential, but the mandate is not zero-sum. Investments in developed and emerging markets hedge risks, expand biodiversity impact, and provide insights from global best practices. Domestically, high-impact sectors will be balanced with non-revenue-generating projects such as peatland restoration or forest buyback programs. Effective financing and investment tools will be critical to achieving these outcomes.

Proactive Compliance with EU Regulations

EU policies, including the Biodiversity Strategy 2030 and LULUCF, guide the strategy from the outset. Forestry assets face increasing risks from climate change and evolving regulations. Early deployment of capital helps the sector adapt, ensuring compliance and readiness for new rules. Proactive planning will enable us to have timely interventions that maximize both ecological and financial outcomes.

Quantifying Our biodiversity impact

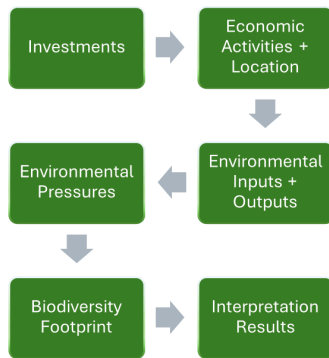
We aim to quantify the Fund's impact on the planet using biodiversity metrics that are scientifically robust and recognized by the financial sector, ensuring high reporting standards. BWF will implement the Biodiversity Footprinting for Financials (BFFI) framework to assess and monitor the portfolio's impact on biodiversity.

The BFFI framework allows tracking biodiversity impacts across the diverse asset classes in which the Fund invests. It aligns with the Fund's dual mandate by relying on data from the EXIOBASE database to evaluate biodiversity impacts and is one of the leading frameworks for financial institutions.

This approach enables the Fund to assess the biodiversity impact of its investments, from ecosystem dependencies at the site level to broader activities, and to account for emissions and biodiversity effects across the entire value chain, including upstream and downstream Scope 3 emissions, using a methodology adapted to the forestry sector.

The framework is suited to the Finnish forestry sector, which is BWF's primary focus. The PDF biodiversity metric within BFFI measures annual species changes per hectare, allowing the Fund to complement forest carbon sink tracking with ecosystem preservation and establish measurable biodiversity targets alongside emissions objectives.

The Fund will manage selected assets to support biodiversity assessment and monitoring and will use complementary tools such as IBAT, which contributes to the BFFI framework and enhances the Fund's capacity for comprehensive biodiversity analysis and tracking.



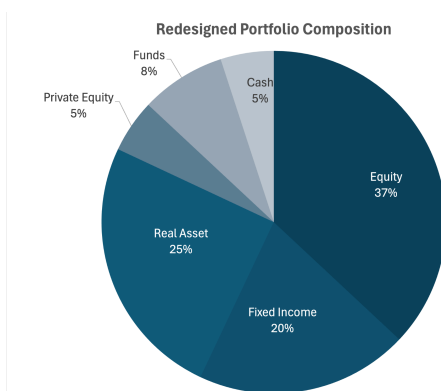
The main steps in Biodiversity Footprinting for Financials (BFFI)

Redesigned Portfolio

Our primary tool for achieving biodiversity goals is the capital we manage. We must redesign our allocation to sustain near-term performance while pursuing our long-term mission. We aim to balance high-yield investments that support biodiversity with underperforming but ecologically critical assets.

Equity remains the core instrument, with 40% of the portfolio allocated to public and private firms, down from 55%, reflecting a shift toward more targeted, active engagement through other asset classes.

Fixed-income instruments, including government bonds, will align with our long-term commitments, while financing and credit will be refocused to channel capital into biodiversity conservation. We will also invest in external funds to hedge risk and learn from leading practices.



Our New Portfolio Allocation

This strategy targets approximately 6.3% annualized returns with 8% volatility, achieving stronger risk-adjusted performance while restoring the ecological assets that support Finland's economy.

Public Equity

The equity asset class, particularly investments in the Finnish forestry sector and related by-products such as biofuel, offers a unique opportunity to shape the nation's forests as a carbon sink by 2035. We will screen companies to assess their current environmental impact and determine the best path toward sustainability.

Our strategy combines financial returns with net positive environmental impact, including emissions reduction and biodiversity preservation. To evaluate companies, we analyze disclosures required by the EU Taxonomy and CSRD, we assess both the sustainability of activities and their net climate impact. We calculate net emissions as the difference between CO₂ emitted and removed, with negative values representing a positive impact. Through the BFFI, ecosystem effects are measured in hectares restored using the PDF model, with targets updated annually. By giving greater weight to net emissions, we identify equities that support or hinder Finland's goal of restoring its forests as a carbon sink.

Applying this to the Finnish forestry sector, UPM, the market leader controlling over half the sector's value, removed more than 4 million tonnes of CO₂ in 2024.⁵ Smaller companies added over 1 million tonnes of net emissions. With Finland's forests netting 10 million tonnes of CO₂ in 2023, this analysis guides our approach. BWF will support UPM and other positive-impact companies by acquiring stakes to follow developments and invest in projects that strengthen the forests' role as a carbon sink. We will also engage with companies currently generating negative emissions, using our capital and active shareholder role to guide their transition toward net positive impact across all scopes.

For non-forestry Finnish equities and global equities, we will rely on the BFFI framework and international

datasets to monitor emissions and biodiversity impact. Using company reports and the PDF metric, we will track changes over time. In these cases, our focus will be on supporting sustainability goals while maintaining a smaller activist role, prioritizing high returns from companies with strong climate disclosure and alignment with global biodiversity standards.

Fixed Income

We increase fixed-income exposure from 15% to 20%, recognising that predictable cash flows are essential for meeting near-term spending obligations and buffering portfolio volatility. The allocation spans Finnish, EU, North American, and emerging-market issuers.

Priority will be given to green bonds and other sustainable instruments that directly finance projects supporting Finland's national net-zero objectives, particularly in forestry and land-use sectors. While returns may be lower, these investments allow the fund to play a strategic role in the country's ecological transition.

Internationally, approximately 15% of fixed-income capital will be allocated to emerging markets. This diversifies the portfolio while channeling funds into nature-positive projects, such as reforestation in high carbon-sequestration regions.

In addition, fixed-income instruments, particularly from North American issuers with strong credit quality, act as a store of wealth over the long term. They provide capital preservation and stability, ensuring the fund maintains financial strength while continuing to advance its biodiversity and climate mandate.

Real Assets

Over 50% of BWF's allocation to real assets will be directed toward forestry, reforestation, and peatland restoration projects. These holdings allow the Fund to act as an active steward of land and biodiversity, leveraging BFFI metrics and complementary tools to track long-term improvements in ecosystem quality.

Continuous monitoring ensures that projects deliver verifiable biodiversity gains alongside carbon sequestration outcomes.

We consolidate the legacy portfolio's green infrastructure and timberland holdings into a natural capital sleeve, allocating approximately 25% of the portfolio to four targeted strategies:

- **Finnish Restoration Forestry (~10%):** Invests in continuous-cover forestry, selective logging, and extended rotation cycles. Short-term yields may fall 15–30%, but these practices regenerate old-growth timber, enhance biodiversity, and generate sustainable revenue through certified timber, premium pricing, and future carbon credits.
- **Finnish Peatland Restoration (~6%):** Funds rewetting and reforestation of degraded peatlands, reversing the trend that turned Finland's forests from a carbon sink into a carbon source. Immediate cash flows are limited, but long-term ecological value and carbon and biodiversity credits are realized.
- **Global Nature Assets (~8%):** Targets biodiversity-rich projects outside Finland, including tropical reforestation, mangrove restoration, and conservation concessions. These can be monetized through carbon offsets and resilience services, supporting sustainable forestry across diverse biomes.
- **Grid, Renewables, and Enabling Infrastructure (~6%):** Supports renewable energy and transmission to reduce pressure on Finnish forests. Projects include grid modernization, distributed solar and storage, and nature-based engineering solutions such as living shorelines.

This real-asset strategy directly advances BWF's mandate to restore domestic ecosystems while

promoting global biodiversity innovation. By blending domestic restoration with global nature projects, we balance local stewardship with higher biodiversity impact abroad. Real assets also provide inflation protection and relatively predictable, uncorrelated returns over long horizons, aligning ecological and financial objectives.

Private Equity and Biodiversity Funds

We allocate a minor fraction to biodiversity-focused funds managed by experienced global players, including impact funds, blended-finance vehicles, biodiversity-themed ETFs, and private equity funds targeting ecosystem innovation. These vehicles invest across restoration projects, carbon markets, nature-positive enterprises, and early-stage companies developing tools for ecosystem restoration such as remote sensing, biodiversity software, cultured timber, bio-based materials, and waste recycling.

Funds like the AXA Impact Fund demonstrate that private capital can support carbon capture, reforestation, and mangrove projects while generating financial returns. By investing in these funds, BWF benefits from specialist expertise, global deal flow, and portfolio flexibility. Early-stage companies require patient capital and longer holding periods, but their potential for outsized returns offsets illiquidity.

Biodiversity Measurement and Standardization

Investing in these funds also supports biodiversity measurement. Many fund managers already track ecological outcomes, offering opportunities to collaborate on standardized metrics and strengthen their adoption within the financial sector. This positions BWF as a leader in advancing transparency and accountability in conservation finance.

Voluntary Carbon Market (VCM) Exposure

These asset classes provide exposure to the Voluntary Carbon Market, which increasingly targets high-yielding projects such as Afforestation, Restoration, and Revegetation (ARR) and mangrove restoration. These projects deliver both carbon removal and biodiversity

benefits. With the VCM currently valued at \$2 billion and projected to grow into the hundreds of billions, early exposure supports both impact and returns. Projects certified under Verra and Gold Standard meet strict biodiversity requirements, including additionality, leakage control, stakeholder consultation, and monitoring.

Engagement in the VCM also supports Finland's net-zero target under Article 6 of the Paris Agreement. By participating in carbon credit trading, BWF can help offset emissions if targets are unmet or generate revenue by selling surplus credits. This allows the Fund to contribute directly to Finland's climate leadership while achieving financial and ecological objectives.

Strategic Value

Our redesigned portfolio positions the Borealis Wealth Fund as a global leader in biodiversity investing. Capital is directed to high-quality assets with clear ecological and financial value. Compared to the legacy portfolio, this allocation broadens diversification, embeds biodiversity across asset classes, and creates a pipeline of innovative projects.

Liquidity Management Plan

We will maintain 5% of the portfolio in cash to cover operational needs and unexpected shocks, capped at 10% to balance security with return potential. Short-term government bonds will complement this buffer, preserving capital while generating moderate returns. Liquid public equities provide flexibility for rapid reallocation, while illiquid assets such as forestry and restoration projects are held longer to achieve biodiversity goals and durable returns.

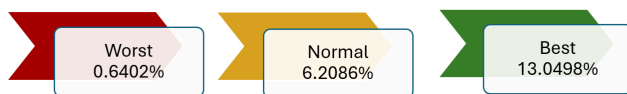
Seasonal cash-flow needs, driven by planting, harvesting, and permitting cycles, will be met by temporarily increasing the cash buffer to around 12% during peak periods, ensuring strategic allocation and ecological commitments remain uninterrupted.

Risk Analysis

It is important to safeguard our financial assets to support our dual mandate. Scenario models and stress tests show how the portfolio responds to shifts in market, climate, and regulatory conditions.

Results highlight that in supportive environments, where policies remain stable and biodiversity markets grow, the portfolio performs strongly and channels more capital into restoration and low-impact sectors. Under severe stress, with prolonged timber yield shocks, falling carbon prices, and intensified climate-driven damage such as wildfires and drought, returns weaken and the pace of forest restoration slows. Liquidity in public equities and the cash buffer provide stability, protecting capital and keeping core biodiversity projects funded even in challenging periods.

We also find that transition and stranded-asset risks are concentrated in companies and sectors that fail to adapt to tightening EU and Finnish biodiversity and emissions rules. Identifying these vulnerabilities early enables us to adjust allocations, reduce long-term losses, and push for sector-wide transformation.



Estimated annual returns under different scenarios

Implementation Plan

We will start rolling out our new allocation plan without delay, while honoring our existing commitments in infrastructure and towards private equity, making divestments if they no longer align with our biodiversity objectives. When it comes to public equity, the Fund's adjustments will be phased to minimize losses, negotiating premiums where possible.

Priority is given to forming partnerships with key stakeholders. The Finnish government, local forestry agencies, restoration funds, and global biodiversity initiatives like the METSO will play a central role.

Reallocation of weights

BWF will actively adjust portfolio weights across asset classes, geographies, and sectors to maximize biodiversity impact and financial returns. Initially, about 50% of equities will target leading innovators to support sustainable technologies. Over 3–5 years, allocations will expand to lagging firms, guiding adoption of better practices and forming global partnerships.

We will divest from the bottom 10% of underperforming assets once performance insights are clear. Returns from early investments will fund historically low-performing sectors, such as peatland restoration or forest buyback programs, initially 5–10% of total capital. Quarterly reviews of IRR, biodiversity scores, carbon sequestration, and regulatory alignment will ensure reallocations strengthen both ecological outcomes and portfolio resilience.

Measurements Enhancing

Through the BFFI tool, BWF will access scientifically robust databases such as EXIOBASE, integrate additional satellite-based forest monitoring, habitat connectivity indices, and soil carbon metrics to track biodiversity and financial performance. We will combine remote sensing, company disclosures, and field verification in partnership with NGOs. Third-party audits, including FSC validate results annually.

This ensures that the fund keeps capital allocation dynamic, prioritizing sectors and regions with the highest biodiversity impact per euro invested.

Acting Today for Lasting Impact

Finland's forests are at a critical juncture, vital to the nation's net-zero target. BWF can act now to protect and restore these assets. Our strategy demonstrates that financial returns and ecological stewardship can go hand in hand. The key message is clear: proactive, informed action today secures Finland's natural wealth for future generations while generating durable financial returns.

Appendix

Appendix 1: Legacy Portfolio Recreation

Using appropriate ETFs as proxies for each asset class, we reconstructed the historical performance of the Legacy Portfolio. The equity allocation was divided into three major geographic regions (Europe and Finland, North America, and Emerging Markets) represented respectively by the Nuveen ESG International Developed Markets Equity ETF (NUDM), Vanguard ESG North America All Cap UCITS ETF (V3NB), and Nuveen ESG Emerging Markets Equity ETF (NUEM). A composite equity portfolio was then derived by taking the weighted average of these three ETFs, using regional weights of 0.85, 0.10, and 0.05. For the remaining asset classes, we used the iShares € Green Bond UCITS ETF (GRON) to proxy green bonds, the iShares Global Clean Energy ETF (ICLN) for green infrastructure, and the iShares Global Timber & Forestry ETF (WOOD) for forestry and land holdings. The overall performance of the Legacy Portfolio was estimated as the weighted sum of each asset class’s return, consistent with the fund’s strategic asset allocation.

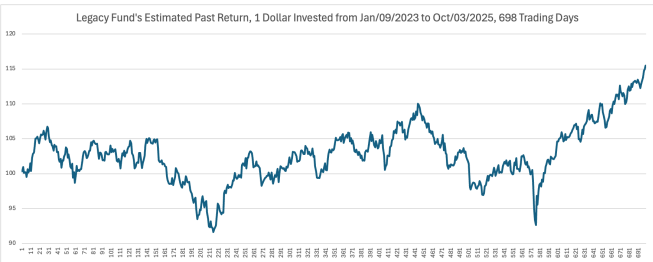
Based on the recreated performance of the Legacy Portfolio, the ESG-filtered equity allocation delivered the strongest returns, with an average annual return of 14.27% and a volatility of 14.85%, highlighting its role as the primary growth driver. Green bonds provided stability with a 1.74% return and 5.97% volatility, serving as a defensive anchor within the portfolio. In contrast, green infrastructure underperformed, recording an average annual return of -6.72% with the highest volatility at 23.70%, reflecting the sector’s sensitivity to clean energy market fluctuations. Forestry and land holdings generated a modest 0.49% return with 17.83% volatility, consistent with the cyclical nature of timber and land assets. Finally, the cash allocation, proxied by the 3-Month EURIBOR, contributed a 3.21% return at an exceptionally low 0.11% volatility, providing liquidity and stability to the overall portfolio.

The recreated Legacy portfolio has an average annual return of 6.04% with **Shape ratio of 24.27%**.

The correlation matrix of each broad asset class is also estimated using the same approach.

Correlation Matrix					
	Green Bonds	Green Infra	Forest&Land	Equity	Cash
Green Bonds	1.0000	-0.0567	-0.0250	-0.0356	-0.0350
Green Infra		1.0000	0.5881	0.6016	0.0132
Forest&Land			1.0000	0.7158	0.0554
Equity				1.0000	-0.0461
Cash					1.0000

Recreated portfolio compounded return post-COVID



Appendix 2: Expected Annualized Returns and Volatility for Biodiversity-Focused Asset Classes (2025–2034)

Biodiversity-directed investment assets have only recently gained market prominence. Given their short performance history and the extraordinary volatility of the COVID-19 period, historical data alone provides limited insight into their long-term return potential. Consequently, part of the return estimation relies on forward-looking reasoning and evidence drawn from relevant academic and industry literature, rather than purely empirical backtesting.

The table below summarizes estimated nominal annualized returns and standard deviations for each asset sleeve in the redesigned portfolio. Values reflect a mid-point of plausible ranges and are expressed in EUR. They do not account for currency hedging and assume a moderate inflation environment (~2 %).

Asset Class (Biodiversity Focused)	Expected Annual Return (10-yr)	Expected Volatility (σ)	Rationale & Sources
Biodiversity-screened public equities			
– Finnish biodiversity equities	7.5 %	20 %	Long-term Finnish stock returns average ≈ 10.95 % with ≈ 21 % volatility . Screening for biodiversity likely removes environmentally destructive companies but does not materially change beta. I therefore project a 7.5 % return (slightly below the historical average to reflect lower growth) with ~ 20 % volatility.
– EU biodiversity equities	6.7 %	15 %	BNY's CMA for non-U.S. developed equity expects 6.7 % return with 16.3 % standard deviation . Using EU economic and policy headwinds, I assume a similar return and slightly lower volatility because biodiversity-leaders are often larger, more stable firms.
– North American biodiversity equities	7.5 %	16 %	U.S. equity CMAs call for a 7.5 % return and 16.2 % volatility . Biodiversity screens should have minimal drag and may reduce certain tail risks (litigation, regulatory penalties).
– Emerging-market biodiversity equities	8.0 %	19 %	BNY's expectation for emerging-market equities is 7.7 % return with 18.7 % volatility . EM biodiversity companies may capture growth in nature-positive products; I add a small premium to 8 % while acknowledging higher volatility (~ 19 %).

Biodiversity-screened fixed income			
– Finnish biodiversity bonds	4.5 %	5 %	Finnish government and corporate bond yields benefit from euro-area monetary easing but remain modest. BNY projects U.S. aggregate bonds at 4.8 % with 5.1 % vol ; adjusting for lower Finnish yields and shorter duration, I estimate 4.5 % return and 5 % volatility.
– EU biodiversity bonds	4.2 %	6 %	Global aggregate bonds ex-U.S. are expected to return 3.2 % with 8.3 % vol . EU green/biodiversity-linked bonds typically trade with a small “greenium” over conventional bonds; I assume a slightly higher 4.2 % return and 6 % volatility due to mixed credit quality.
– North American biodiversity bonds	5.0 %	7 %	Combining U.S. aggregate (4.8 %) and U.S. high-yield (6.0 %, 8.4 % vol) assumptions , biodiversity-linked coupons and performance-step features could boost returns. I assign a 5 % expected return with 7 % volatility.
– Emerging-market biodiversity bonds	6.5 %	9 %	Emerging-market local-currency debt is forecast to return 4.0 % with 9.0 % volatility . Biodiversity-linked structures may add yield and deliver carbon/biodiversity credit premiums; I project 6.5 % return with similar 9 % volatility.
Real assets			
– Domestic restoration forestry (continuous cover & mixed species)	6.5 %	12 %	BWF plans for IRRs of 5–7 % (8 % with biodiversity credits). Timberland historically produced ~8.72 % annual returns ; combining these and Finland's lower timber prices yields a 6.5 % return with ~12 % volatility (reflecting weather and commodity risk).
– Domestic peatland restoration SPVs (blended finance)	6.0 %	7 %	Project IRRs in the BWF strategy range 4–6 % base and up to 9 % when carbon credit markets mature. Assuming moderate uptake, I set 6 % with 7 % volatility, given revenue streams from carbon/biodiversity credits and low correlation with public markets.

– Global nature assets (certified tropical forestry, mangroves, blue carbon)	8.5 %	10 %	Global timber and blue-carbon projects earn 7–11 % returns according to the BWF plan. Timberland's 8.72 % historical return supports a mid-point of 8.5 %. Diversification across biomes reduces volatility to ~10 %.
– Grid/renewables & enabling infrastructure	6.5 %	10 %	The Dow Jones Brookfield Global Infrastructure Index delivered a 6.77 % 15-year annualized return . Renewable-grid projects with regulated cash flows and inflation linkage justify a 6.5 % return and ~10 % volatility.
Private equity – Environmental & biodiversity-tech	10.5 %	20 %	BWF targets ≈11 % IRR for biodiversity-tech PE. BNY projects 9.7 % returns with 20.1 % volatility for U.S. private equity . Innovation in ecosystem monitoring and nature markets could warrant a small premium (10.5 %) but risks remain high (≈20 % σ).
Other biodiversity funds	6.0 %	9 %	Funds investing in diversified natural-capital projects should achieve returns between green bonds and real assets. Using the balanced portfolio projection of 6.3 % , I assign 6 % with 9 % volatility.
Cash / liquidity buffer	3.5 %	0.5 %	Three-month EURIBOR was around 3.6 % in 2024 according to the BWF strategy. Cash yields are assumed to average 3.5 % over the coming decade with minimal volatility.

Important evidence-based inputs:

1. **Forward-looking capital-market assumptions from institutional managers.** BNY Mellon's 2025 Capital Market Assumptions (CMA) provide expected returns and standard deviations for U.S. equity, non-U.S. developed equity, emerging-market equity, aggregate fixed income, high-yield credit and emerging-market local currency debt . These assumptions incorporate current starting valuations, interest-rate expectations and earnings growth (including AI productivity gains) and therefore offer a credible base for long-term projections.

2. **Historical and index data for natural-capital assets.** Research on timberland and farmland returns shows that the NCREIF Timberland Index delivered an average annual return of 8.72 % from 1992–2023, while farmland returned nearly 11 % annually with a standard deviation around 6 % . These figures were used to benchmark forestry and peatland restoration projects.
3. **Market-specific return and risk statistics.** The Finnish stock market's long-term average return (1988–2021) is ~10.95 % with a stock-price volatility of ~21 % . These data points were used to calibrate Finnish biodiversity equity estimates relative to the global CMAs.
4. **Infrastructure index performance.** The Dow Jones Brookfield Global Infrastructure Index has delivered a 15-year annualized return of 6.77 % . This informed expected returns for grid/renewable infrastructure projects.
5. **Green and sustainable bond market characteristics.** A factsheet for the Northern Trust Global Green Bond Index Fund reports a yield-to-maturity of ~3.4 % and a current yield around 2.8 % , with an average duration of 6.5 years . BNY's CMAs project U.S. aggregate bonds to return 4.8 % annually with 5.1 % volatility and global ex-U.S. bonds to return 3.2 % with 8.3 % volatility .
6. **Portfolio design guidelines from the BWF strategy.** The fund's own strategic plan describes expected internal rates of return for restoration forestry (5–7 %), peatland SPVs (4–6 % base with upside to 9 %), global nature assets (7–11 %), green infrastructure (≈6.2 %), and biodiversity-tech private equity (≈11 %). These benchmarks were blended with market data to derive return distributions.

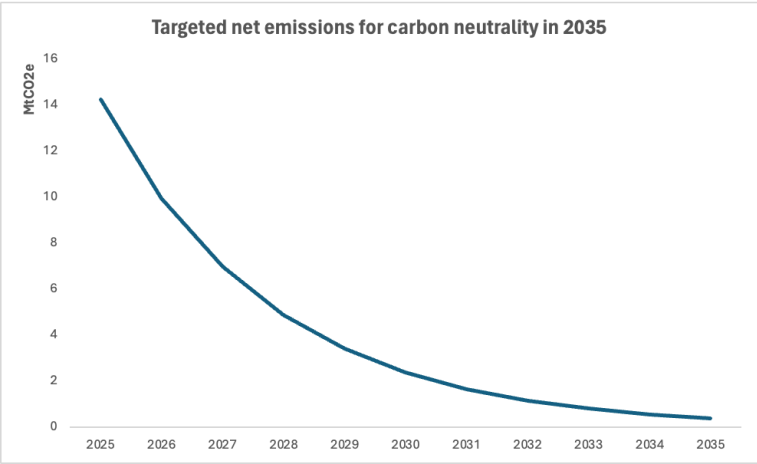
Where academically reviewed studies of biodiversity-specific funds were unavailable (e.g., peatland restoration vehicles), I anchored estimates to adjacent asset classes (timberland, green bonds, infrastructure) and applied conservative assumptions consistent with the risk profiles described in the BWF strategy. In practice, biodiversity screens often have neutral to mildly positive performance effects because they mitigate environmental liabilities; I therefore did not assume a significant return discount relative to the underlying asset classes.

Based on the recreated performance of the Legacy Portfolio, the ESG-filtered equity allocation delivered the strongest returns, with an average annual return of 14.27% and a volatility of 14.85%, highlighting its role as the primary growth driver. Green bonds provided stability with a 1.74% return and 5.97% volatility, serving as a defensive anchor within the portfolio. In contrast, green infrastructure underperformed, recording an average annual return of -6.72% with the highest volatility at 23.70%, reflecting the sector's sensitivity to clean energy market fluctuations. Forestry and land holdings generated a modest 0.49% return with 17.83% volatility, consistent with the cyclical nature of timber and land assets. Finally, the cash allocation, proxied by the 3-Month EURIBOR, contributed a 3.21% return at an exceptionally low 0.11% volatility, providing liquidity and stability to the overall portfolio.

Appendix 3: Estimation of the Forestry Sector Market Capitalization and level of CO2 emitted by its companies

Using the fact that the Forestry Sector represented 3.7% of GDP in Finland in 2020 according to the case, we estimated the market capitalization of Finnish forestry companies relative to the total Market Capitalization. Collecting data from market leaders in the Finnish forestry sector, such as UPM, Metsa Group, Stora, we computed the ratio of net emissions (emissions less removals of CO2) relative to the market capitalization of the company. Weighting these ratios according to the share of each company within the forestry sector, we estimated the average net emissions for companies for which we didn't collect the data for, to then estimate the net emissions produced by the whole sector.

This yielded net emissions of more than 14 million tonnes of CO2, which corresponds to the levels from the Exhibit given in the case showing the levels of emissions generated by the Finnish forest as a net carbon source in 2023: 12.5 MtCO2e. From that graph, we find that a representative company would have to decrease on average their net emissions by 30% every year to reach the sector's net zero emission target that we set.



Appendix 4: Key Environmental Regulations and Portfolio Implications

The table above provides an overview of the key regulatory frameworks shaping Finland's forestry, land-use, and emissions landscape. It highlights each regulation's purpose, mechanism, current status, and 2030 targets, as well as tangible examples of how companies and institutions are responding.

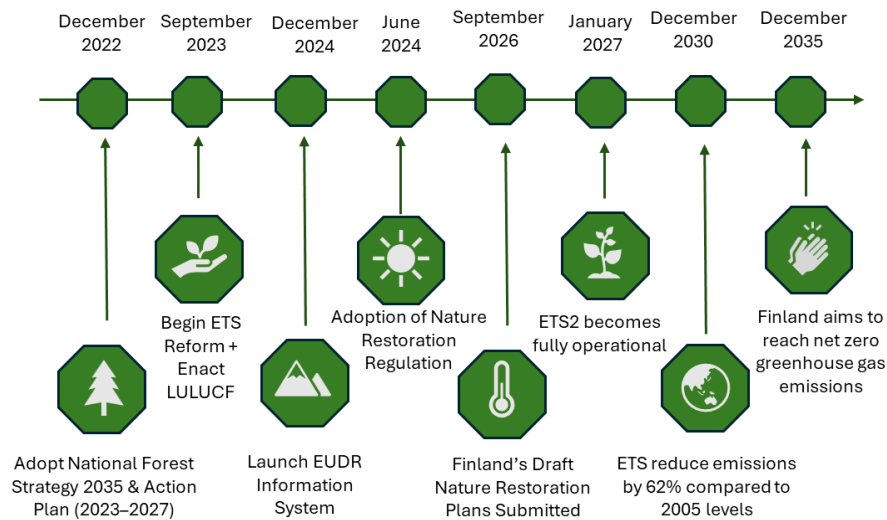
By summarizing the projected annual reductions and ongoing changes, the table contextualizes the challenges and opportunities for the Borealis Wealth Fund. It underscores how regulatory alignment, strategic capital deployment, and active engagement with market participants can drive both compliance and measurable ecological impact.

This framework allows BWF to identify sectors and assets where investment can accelerate progress toward net-zero and biodiversity objectives, while also providing insight into potential risks from underperforming or non-compliant areas.

Regulatory Impact Table

Regulation	Purpose	Mechanism / Effect	2030 Target	Current (2024)	Change from 2023. Annual Reduction Needed	Tangible Examples / Notes
LULUCF(Land Use, Land-Use Change, Forestry)	Ensure land use acts as a net carbon sink to meet climate targets	Binding EU-wide target of 310 MtCO ₂ e net removals by 2030; Finland aims for 17.8 MtCO ₂ e by 2030 via sustainable forest management, peatland restoration, and long-lived wood products	17.8 MtCO ₂ e net removals (Finland) by 2030	Finland's LULUCF sink was about 3 Mt in 2022, aiming to increase to 17.8 Mt by 2030. Current emissions from LULUCF rising, with sink historically fluctuating and weakened recently	Sink fell about 6% from 2023 (exact 2024 data varies: net sink under pressure due to high felling). Approx. 9% annual increase in net removals needed	UPM engages in LULUCF-aligned management and peatland restoration. The Finnish government's Land Use Climate Plan supports these targets
EU ETS(European Union Emissions Trading System)	Cap and trade system for major emitters; key tool for 55% EU GHG reduction vs 1990 level by 2030	Emissions cap declines annually by 4.3%-4.4%. Emissions trading for energy, industry, aviation sectors.	~2.52 GtCO ₂ e EU-wide; Finland's share varies	EU ETS emissions fell 5% in 2024 overall, driven by power sector decarbonization. Finland's ETS emissions approx. 13.3 Mt in 2024, down 13.3% from 2023	Approx. 11% annual reduction needed aligning with EU goals	Neste Oyj reducing emissions and purchasing allowances. EU ETS seen as effective in emission reduction
ESR(Effort Sharing Regulation)	Covers non-ETS sectors: transport, buildings, agriculture, waste; Finland must reduce ≥50% by 2030 vs 2005	National binding targets with policies like fuel taxes, biofuel mandates, EV subsidies	50% reduction (Finland) by 2030 vs 2005	Predicted 2% annual reduction; 2024 emissions around 6.9 Mt, decreased ~1% from 2023	Approx. 2% annual reduction needed	Government incentives for EVs, biofuels. Posti Group electric fleet example
COP15 30x30 target	Global commitment to conserve 30% of Earth's land and oceans by 2030 to halt biodiversity loss	It aims to safeguard 30% of global land and 30% of oceans by 2030, protecting over 60 million km ² of ecosystems and countless species.	Estimation: 30% of global land could help avoid ~11B tonnes of CO ₂ / yr	Internationally, 37.8 billion tonnes (Gt) of CO ₂ was produced this year. Emissions increased from last year's 37.4B.	Approx. 19% annual reduction needed to achieve goal	Unilever is implementing the COP15 30x30 goals by sourcing all key agricultural materials sustainably and restoring 1.5 million hectares of land, forests, and oceans by 2030. Which is in line with the Kunming–Montreal Framework.

Timeline showing implementations of key regulations and framework:



Appendix 5: Biodiversity metrics used by BWF

Biodiversity Measurement Tool	Description & Use	Data used	Biodiversity Pressures Screened	Assets & covered
BFFI	Measures potential biodiversity loss in PDF per ha per year, identifying dependencies on different ecosystems, as well as evaluating net impacts of economic activities, yielding a measure of the hectares of land affected positively or negatively Commonly used across financial institutions targeting biodiversity solutions	EXIOBASE, which takes into account economic activities as well as world trade flows ENCORE and IBAT methodologies as inputs for additional information	Scope 1 to 3 emissions (although less for Scope 3 downstream) Land use change, Climate Change PDF - leading metric for change in species on a hectare of land	Corporate Loans, Equity (Private and Public), Corporate Bonds, Sovereign Bonds, Impact Funds, Green Bonds

IBAT	Identification and Assessment of biodiversity risk opportunities, analyzing exposure of locations and sites to sensitive biodiversity areas and species	WDPA WDKBA IUCN Red List of Threatened Species	Land use change, Invasive Species STAR metric which as a measure of species richness of a hectare	Corporate loans, Impact Funds, Project Finance
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Appendix 6: Stress Testing and Sensitivity Analysis

Our portfolio stress testing employs Monte Carlo simulation with 10,000 iterations to evaluate performance across diverse market, climate, and regulatory conditions.

Metric	Value
Mean Portfolio Return	6.23%
Portfolio Volatility	4.14%
5th Percentile (Worst-Case)	-0.64%
50th Percentile (Median)	6.21%
95th Percentile (Best-Case)	13.05%

- Baseline Scenario (50th-95th percentile outcomes) Returns of 6.2%-13.0% occur when policies remain stable, biodiversity markets expand steadily, and climate impacts align with historical patterns. The portfolio channels maximum capital into restoration and generates strong risk-adjusted returns.
- Moderate Stress (15th-50th percentile outcomes) Returns of 2%-6% reflect temporary policy uncertainty, carbon price volatility, or regional timber yield reductions of 10-15%. Liquidity buffers absorb short-term shocks while core biodiversity projects continue.
- Severe Stress (0-5th percentile outcomes) Returns approaching -0.64% represent combined shocks: prolonged timber yield declines (25-35%), carbon price collapses (50%+ declines), and intensified climate damage from wildfires and drought. Public equity liquidity (35% of portfolio) and cash reserves (10% of portfolio) protect capital and maintain minimum restoration funding.

Stress Test Assumptions

- Forest Assets: Timber yield sensitivity of -0.8% return per 1% yield reduction; carbon price elasticity of +0.4% per 10% price change; non-linear climate damage at extreme weather thresholds.
- Public Equities: Beta 0.85-1.15 to broader markets; ESG premium compression of 5-12% under stress; regulatory sensitivity of +3% to +8% for EU Nature Restoration Law leaders.
- Liquidity Buffer: 8-12% cash allocation earning zero real return but providing immediate capital for commitments and redemptions.
- Climate Variables: Baseline assumes 1.0x historical climate damage frequency; severe stress applies 3.5x multiplier for high-intensity events.

Transition and Stranded Asset Risk

High-Risk Exposures

Concentrated in sectors failing to adapt to EU and Finnish biodiversity regulations:

- Conventional forestry without biodiversity integration (18% probability of 30%+ impairment by 2030)
- Peat extraction operations (45% probability of regulatory shutdown)
- Monoculture timber in designated protection zones

Early Warning Indicators

Quarterly monitoring triggers reallocation when:

- Regulatory compliance scores fall below 60/100
- Three consecutive quarters of biodiversity metric deterioration
- Climate risk exposure exceeds 25% of asset value
- Credible stranded asset threat within 5-year horizon

Dynamic Allocation Response

Under Severe Stress (5th percentile outcomes):

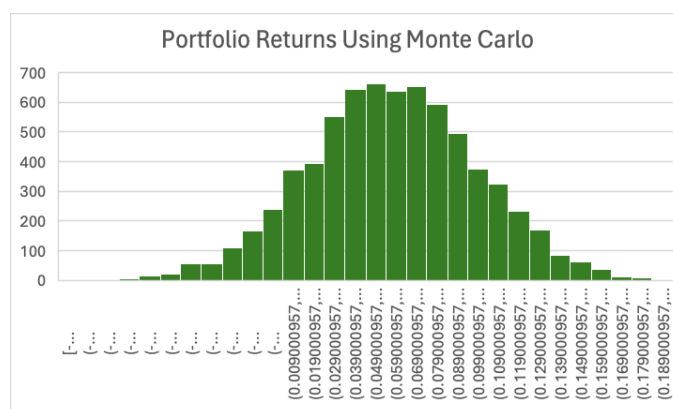
- Increase liquidity buffer from 10% to 15%
- Reduce high-transition-risk sector exposure by 8-12%
- Prioritize diversified forest assets (timber + carbon + biodiversity credits)
- Active engagement on climate adaptation; divest non-responders within 18-24 months

Under Baseline Conditions (median-to-best outcomes):

- Deploy liquidity into distressed biodiversity assets
- Accelerate restoration when land prices decline
- Increase allocation to regulatory leaders

Model Limitations

- Biodiversity market data limited (<10 years); parameters based on analogous markets
- Extreme tail risks may be underestimated due to climate uncertainty
- Policy shift probabilities are qualitative; regulatory changes can be abrupt
- Asset class correlations may strengthen during crises beyond model assumptions



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